KUSHAL YARLAGADDA

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Education

B.Tech, Computer & Communication Engineering, Amrita Vishwa Vidyapeetham — CGPA: 8.43/10
Minor in Data Science, Amrita Vishwa Vidyapeetham — CGPA: 9.1/10
B.Sc. in Data Science (Online), IIT Madras — CGPA: 9.23/10

Experience

Software Engineer I

Apr 2026 – Present

HashedIn by Deloitte, Hyderabad

- Working on **Meridian**, a client-facing enterprise platform using **Angular 21**. Contributing to modular frontend architecture, feature development, and performance optimization across large-scale enterprise modules.

Software Engineer Intern

Jan 2026 – Apr 2026

HashedIn by Deloitte, Bengaluru

- **Developer Experience (DevEx) Platform** *(Python, FastAPI, React, PostgreSQL)*: Built an internal engineering insights platform with GitHub activity tracking, anonymous pulse surveys, and RBAC for Dev / Manager / Admin / Stakeholder roles.
- **GenAI – Product-to-Code Multi-Agent System** *(Python, FastAPI, LangGraph, pgvector, Langfuse)*: Built a multi-agent workflow converting product requirements into Epics → User Stories → Specs → Working Code with pause/resume checkpointing, WebSocket real-time progress, Mermaid diagram generation, and full LLM observability via Langfuse.
- **Home Solutions REST API** *(Java 17, Spring Boot 3, PostgreSQL, JUnit 5)*: Designed a role-based REST API (Customer / Expert / Admin) with JWT authentication, layered architecture, payment lifecycle, expert onboarding/approval flow, and **80%+ test coverage** via JUnit 5 + Mockito with Swagger documentation.
- **HouseMate – On-Demand Service Platform** *(Angular 19, NgRx, RxJS, Karma/Jasmine)*: Built a customer and expert SPA with 5-slice NgRx state, 4-step booking wizard, payment modal (Card/UPI/Net Banking), role-based guards, lazy-loaded routes, i18n (en/hi), and WCAG-AA accessibility; **80%+ unit test coverage** via Karma.
- **IT Help Desk Platform** *(Node.js, Express, React, Jest, Vite)*: Developed a ticket management system with role-based dashboards and real-time status tracking; **80%+ test coverage** via Jest + Vite.
- **Data Engineering – Medallion Architecture** *(Azure Databricks, PySpark, Delta Lake, ADF)*: Built Bronze/Silver/Gold pipelines with SCD Type 1 & 2, CDC, and Star Schema modeling; orchestrated ETL/ELT workflows via ADF and Databricks job clusters.

Skills

- **Languages:** Python, Java, TypeScript, SQL, C
- **Backend:** FastAPI, Spring Boot 3, Node.js/Express, REST APIs, JWT
- **Frontend:** Angular 19/21, React, NgRx, RxJS, HTML, SCSS
- **AI/ML:** LangGraph, LangChain, CNNs, TensorFlow, YOLO
- **Data Engineering:** Azure Databricks, PySpark, Delta Lake, ADF, SCD 1&2, CDC, Star Schema, ETL/ELT
- **Databases:** PostgreSQL, pgvector, SQLite
- **Cloud & DevOps:** AWS, Azure, Oracle Cloud, Git, Docker
- **Testing:** JUnit 5, Jest, Mockito, Karma/Jasmine, Vite

Academic Projects

ECG Anomaly Detection using CNNs *Python, TensorFlow*

- Built a CNN-based ECG classifier for arrhythmia detection using QRS and RR interval features; achieved **97% test accuracy**. Published at IEEE INDISCON 2025.

Medicinal Plant Classification – Handheld Device *MobileNetV2, Raspberry Pi*

- Deployed an offline plant identification system on Raspberry Pi trained on 96 medicinal species. Published at IEEE INCIP 2025.

Biometric Cattle Identification *Xception, Python*

- Developed deep-feature muzzle-pattern recognition with **93% accuracy**; secured **3rd Place** at TANUVAS Digital Veterinary Hackathon 2024.

Research Publications

CNN-Based ECG Signal Analysis for Anomalous Heartbeat Detection
IEEE 6th India Council International Subsections Conference (INDISCON), Rourkela, 2025
DOI: 10.1109/INDISCON66021.2025.11251818

Handheld Device for Medicinal Plant Identification
IEEE International Conference on Intelligent Processing (INCIP), Bangalore, 2025
DOI: 10.1109/INCIP64058.2025.11019308

Certifications

- **Oracle Cloud Infrastructure 2025 Certified Data Science Professional** Oct 2025 – Oct 2027
- **AWS Educate – Machine Learning Foundations**, Amazon Web Services Jul 2025
- **AWS Academy Graduate – Cloud Foundations**, Amazon Web Services Jun 2025
- **IBM – Getting Started with Artificial Intelligence** Jul 2025

Field Research & Social Impact

- Conducted participatory rural research in Assam under **Live-in-Labs**, analyzing infrastructure, agriculture, and livelihood conditions using PRA techniques.